

KRISHNA WIJAYA

GIS Analyst

✉ krishnawijaya@gmail.com

📍 Jakarta, ID

☎ 0878 3901 2249



SUMMARY

A big fan of metalcore. Robust experience in geospatial data management, multicultural teamwork, data driven solutions, and a commitment to ethics. Adaptable, empathetic, and impact oriented. Dedication to team's mission, values, and continuous improvement.

EXPERIENCE

GIS Officer for Rewetting & Revegetation Program

BRGM RI

📅 02/2020 - 11/2024

📍 Jakarta, ID

Provided integrated planning for peatland restoration project with relevant stakeholders in Riau resulting in a measurable restoration target.

- Developed a reporting app using ArcGIS Survey123 for clean and efficient workflows.
- Designed and implemented data engineering solutions for complex spatial datasets handling.
- Conducted a feasibility study for rewetting infrastructure, revegetation, and local livelihood enhancement.
- Provided technical support and FPIC to non-technical communities resulting in increased productivity.
- Collaborated on monitoring, evaluating, and evaluating procedures with relevant teams/stakeholders at project sites.
- Optimized spatial data processing by 40% through innovative data engineering solutions.

Assistant Professional Staff

DCKTRP DKI Jakarta

📅 03/2019 - 12/2019

📍 Jakarta, ID

Provided the updated integration data of land parcel for Jakartasatu platform.

- Enriched land parcel dataset for Jakarta Barat City, resulting in 30% increase in information accuracy for Jakartasatu platform publishing.
- Performed quality control for spatial dataset and legal document attachment finalization.
- Improved land parcel dataset for Jakarta Barat City, resulting in 6000+ spatial records accuracy.

GIS Executive

PT Geo Indo Asia

📅 06/2018 - 02/2019

📍 Yogyakarta, ID

Developed spatial data engineering of road damage information resulting in streamlined workflows.

- Performed inspection of road damage information using ImajView in platform for national road segment in Sumatera Region.
- Developed GIS Toolbox resulting increased in team productivity.
- Provided technical training to new employees for optimal onboarding.

GIS Operator

PT Bhumi Prasaja

📅 05/2017 - 12/2017

📍 Yogyakarta, ID

Provided geospatial based solution for governmental development policy.

- Performed land system suitability analysis for recommended cultivation development.
- Enriched map information by conducting a field survey, ensuring data-collecting from all relevant areas.

EDUCATION

Bachelor of Science in Environmental Geography

Universitas Gadjah Mada

📅 08/2013 - 02/2018

📍 Yogyakarta, ID

STRENGTHS



Strong Problem Solver

Utilized GIS and Data Engineering skills to develop scalable solutions for more efficient workflows.



Effective Communicator

Able to clearly convey technical information to non-technical stakeholders.



Impact-oriented

Diligently ensuring geospatial products meet organization standards and delivered in an understandable term.

SKILLS

GIS Analysis

Remote Sensing

MRV

Spatial Modelling

Drone Mapping

Project Management

Spatial Database

Data Visualization

Scientific Writing

Land Suitability Analysis

FPIC

Hydrology Analysis

ACHIEVEMENTS



GIS Application for Emergency Response

Successfully developed a reporting application used by first responders for quick report and analysis of firespot extinguishing.



Leading Team to Successful Project Completion

Managed a technical team to deliver projects on time. Utilized collaborative workspace to tackle inefficient traditional workflows.



Empower to Elevate

Provided numerous GIS tools and modules for learning and enhanced team productivity. As well as empowered locals with GIS solutions.



Champion of National GIS Competition

Successfully developed a geospatial method for housing suitability mapping. Eliminated other participants from well known universities in 2017.





 [linkedin.com/in/krishna-wijaya-05b10b245](https://www.linkedin.com/in/krishna-wijaya-05b10b245)

Portofolio Essential

Experiences - Projects - Maps

Krishna Aji Wijaya | anhsirkwijaya.github.io/Map-Beyond-Border | krishnawijaya@gmail.com | 0878 3901 2249

Collaborative mapping compilation: From smallholder to certification

Providing understandable & doable mapping method for farmers



Custom attribution form (Avenza Maps)

Area

Name Ed

Suyono

Area >

0.59 ha

Perimeter >

318.32 m

Current layer >

FORM B,FORM B AVENZA,FORM B

[Edit layer](#)

Photos Add

Description Add

Style

Orange area 

Created

5/30/25 at 08:43

Attribute schema 

Beneficiaries Code 

Ketompok 

Deca 

The interface displays the name type and associated properties such as an optional photo, displaying orange tag who is selected and shared between users to ensure the collaboration of data collection.

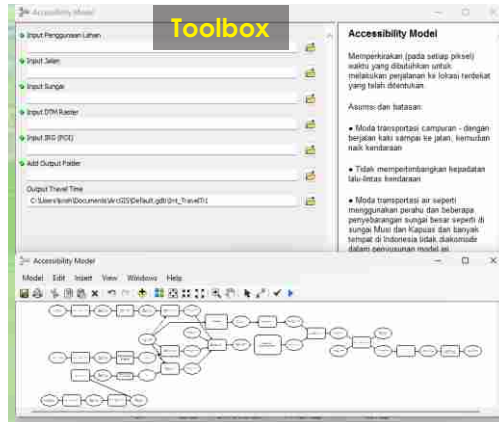
 

- Tackle unnecessary traditional form filling
- Traceable using plot code, timestamp, and geotagged photo
- Multi platform processing
- Educate, empower, & engage the farmer community

Post processing - QC - Compilation



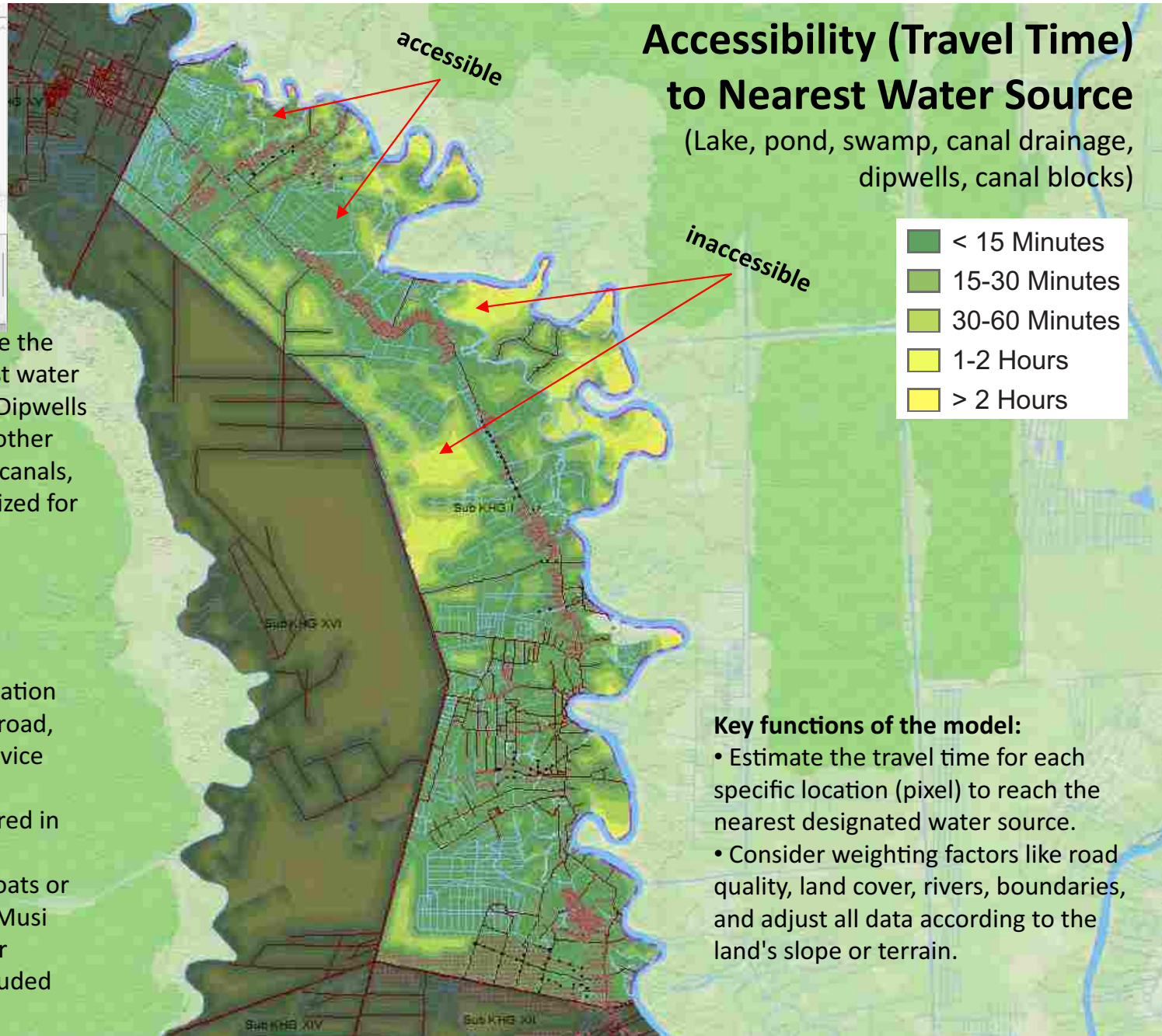
Accessibility Analysis to Nearby Available Water Source During Dry Season (2024)



This model is designed to estimate the time required to reach the nearest water source when a firespot emerges. Dipwells (water pumping spot), as well as other water sources such as reservoirs, canals, and rivers, are assumed to be utilized for fire extinguishing procedures.

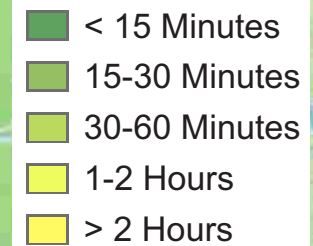
Assumptions and limitations:

- The model uses mixed transportation methods: first walking to reach a road, then using a vehicle to travel and vice versa.
- Traffic congestion is not considered in the model.
- Water transportation, such as boats or large river crossings (e.g., on the Musi and Kapuas rivers, or other similar locations in Indonesia), is not included in this model.



Accessibility (Travel Time) to Nearest Water Source

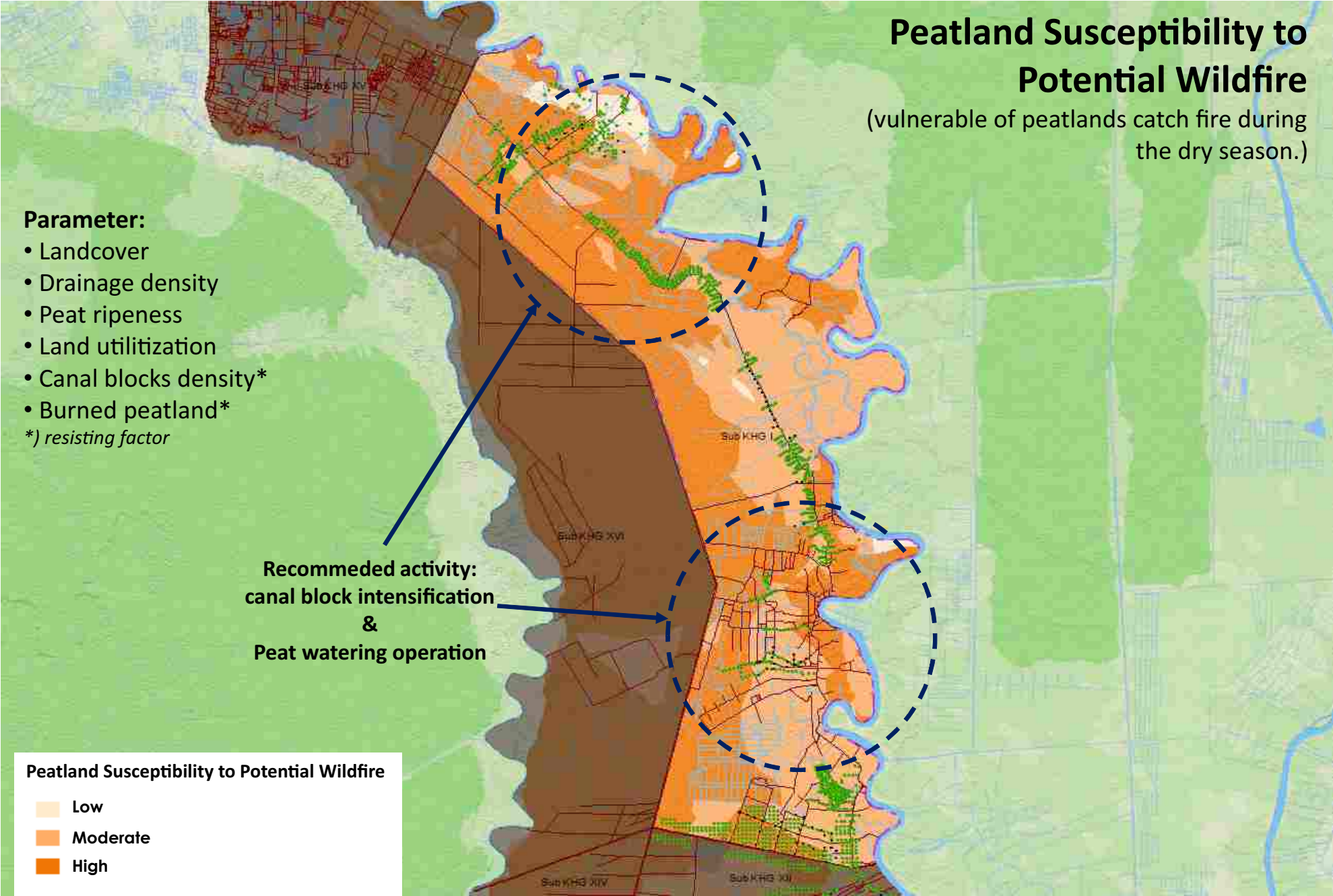
(Lake, pond, swamp, canal drainage, dipwells, canal blocks)



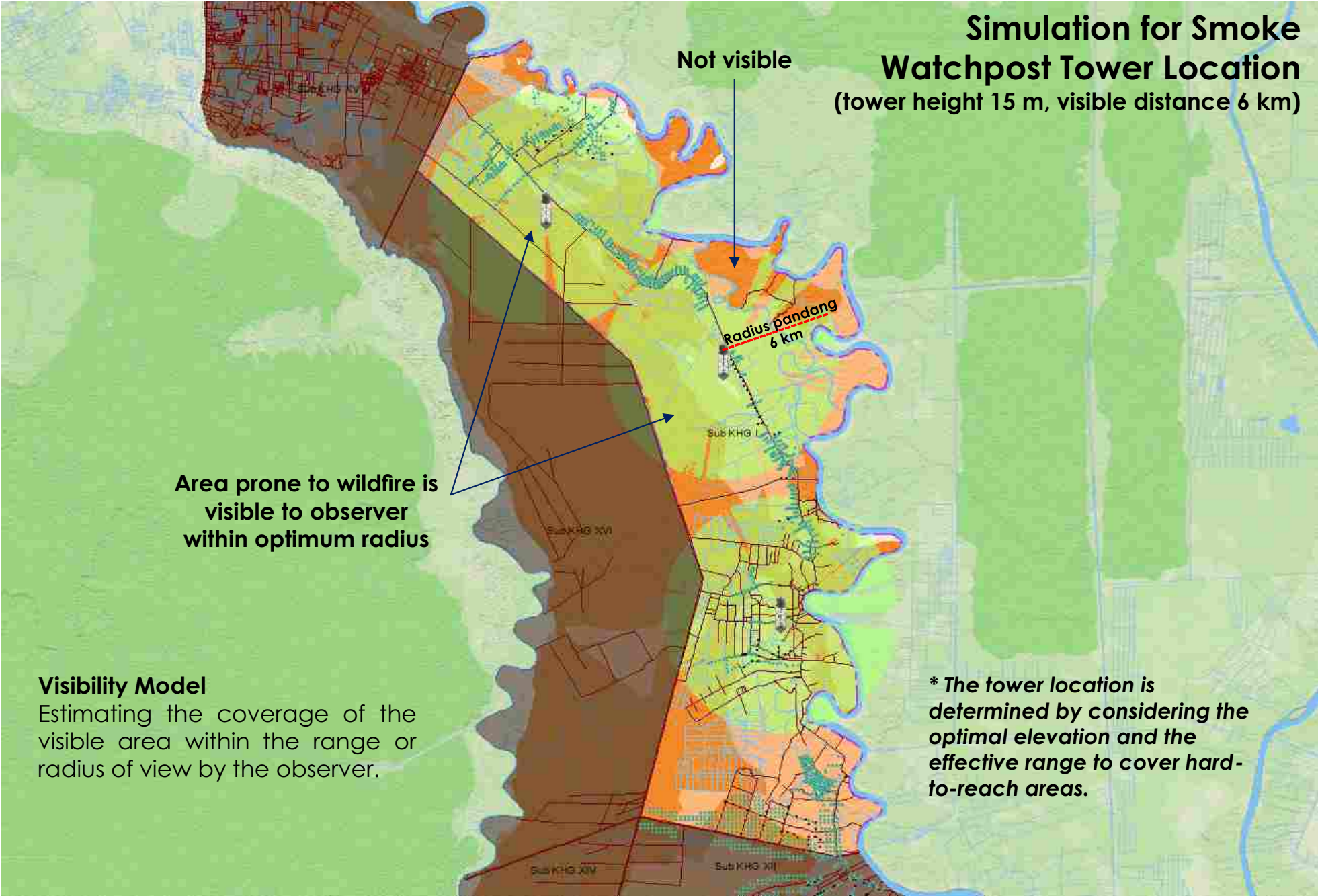
Key functions of the model:

- Estimate the travel time for each specific location (pixel) to reach the nearest designated water source.
- Consider weighting factors like road quality, land cover, rivers, boundaries, and adjust all data according to the land's slope or terrain.

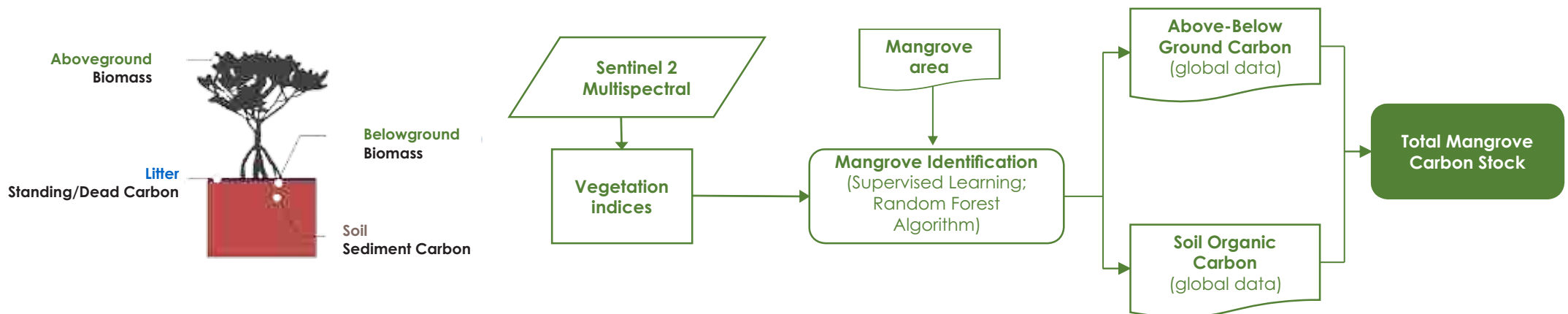
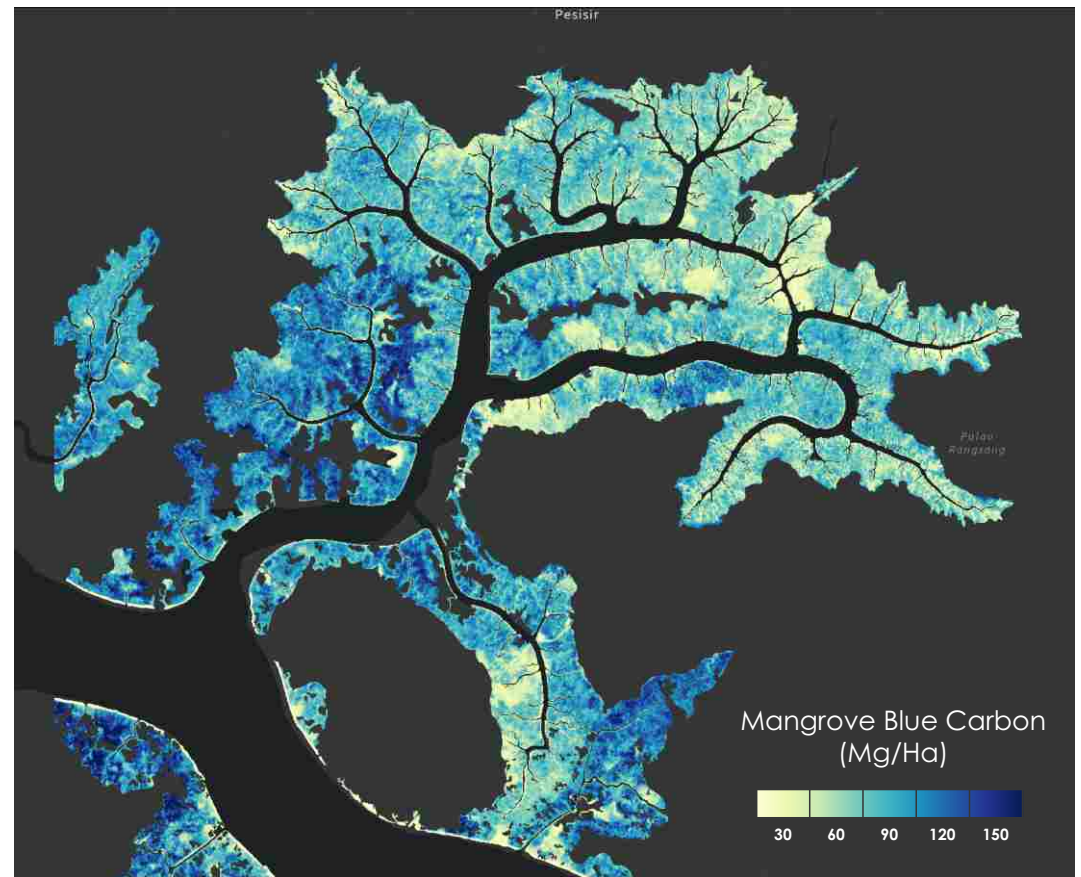
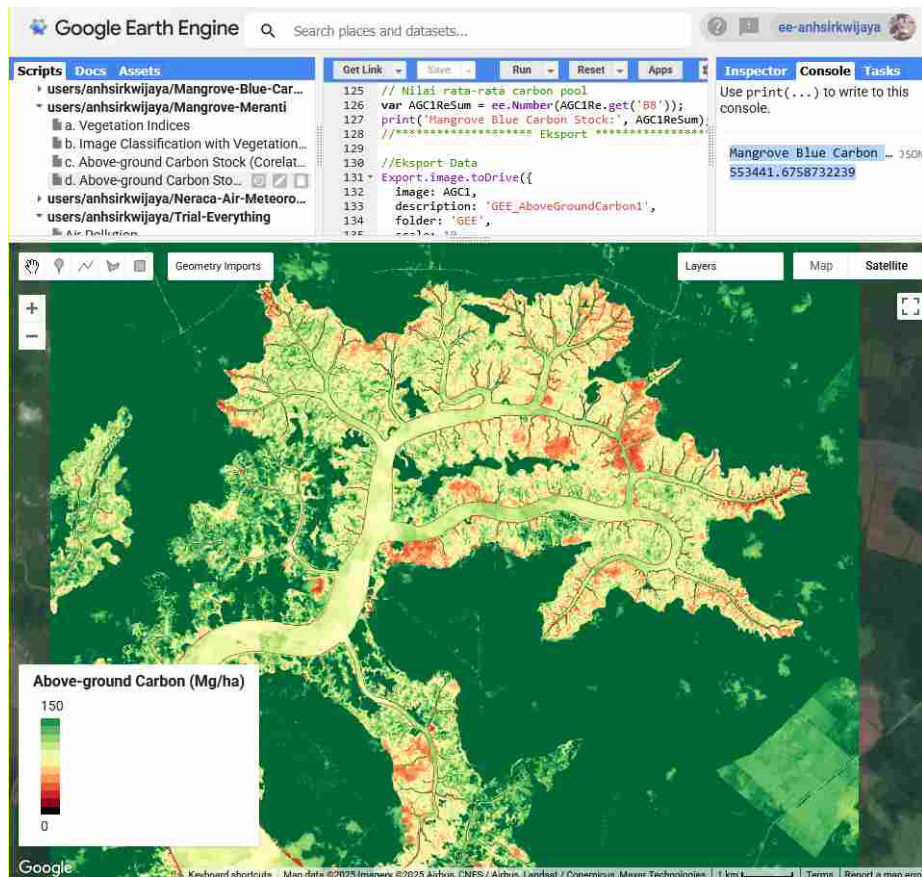
Peatland Susceptibility to Potential Wildfire During Dry Season (2024)



Watchpost Placing Coverage (2024)



Mangrove Blue Carbon Estimation (Below & Above Ground Carbon) 2024



Reporting App for Wildfire Extinguishing Activity using ArcGIS Survey123 (2023)

19:36

Voice LTE1

86%

X

FORM PELAKSANAAN OPLG

Estimasi Luasan yang Dibasahi/Dipadamkan (Ha) *

Angka perkiraan di lapangan

Cuaca *

Arah Angin *

Sebutkan ke arah mana angin berhembus secara dominan

Koordinat *

📍 2°13'S 113°55'E ± 11.6 m

Indonesia

Malaysia

Alat Pembasahan *

☐ Set Mesin Pompa

☐ Pompa Punggung

☐ Peralatan Tangan

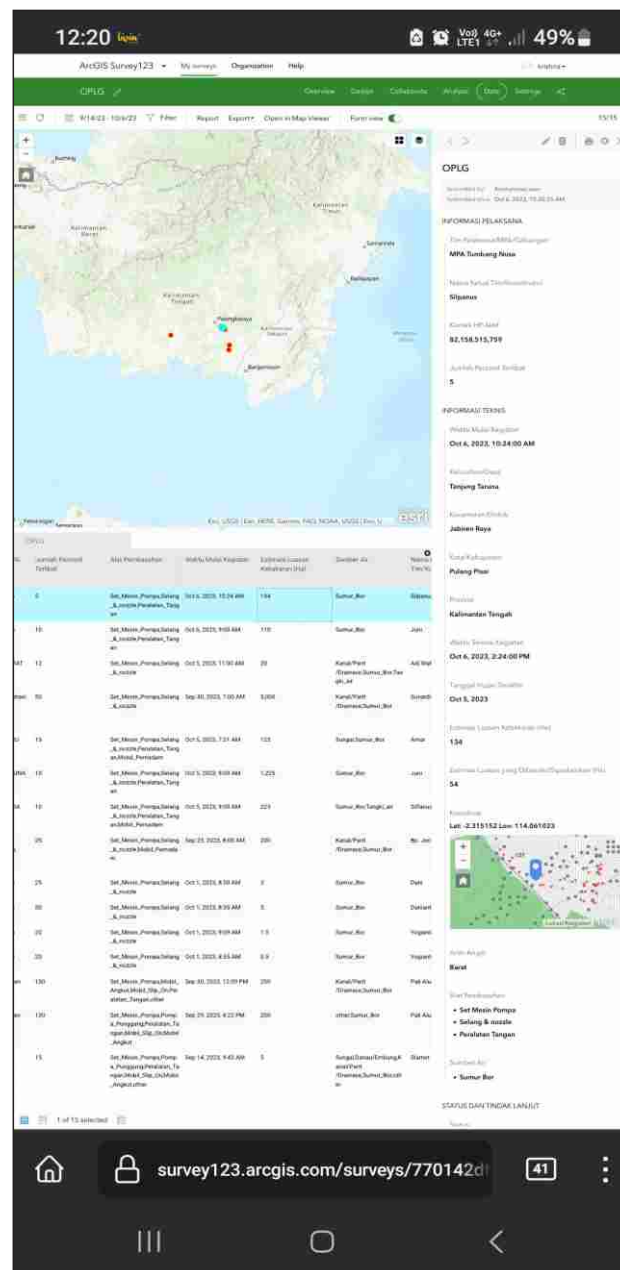
☐ Mobil Angkut

☐ Mobil Slip On (Pemadam)

☐ Lainnya

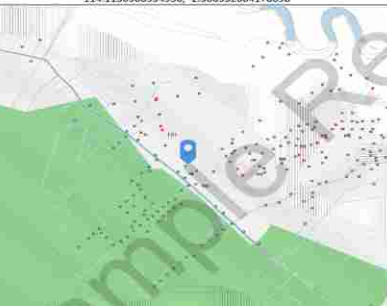
Customizable App Form

offline input --> online submission



Dashboard Map

Live monitoring, accessed from PC & smartphone

 LAPORAN PELAKSANAAN OPERASI PEMBASAHAN LAHAN GAMBUT			
PELAKSANA Pelaksana : MPA Tumbang Nusa Ketua : Junianto No. HP : 081022874456 Jml Personil : 56 Personil Gabungan : BPBO, TNI, Manggala Agni, Masyarakat, Polisi		TANDA TANGAN 	
LOKASI PELAKSANAAN/PEMADAMAN/PEMBASAHAN			
Provinsi	Kalimantan Tengah	Kelurahan/Desa	Tumbang Nusa
Kabupaten/Kota	Pulang Pisau	Kecamatan/Distrik	Jabiren
LOKASI KOORDINAT			
114.1150368534936, -2.366332604178898			
		KHG KHG Sungai Kahayan - Sungai Sebangau HRU 1.03 Karakteristik Gambut Gambut Sangat Dalam (3-5 m) Ekosistem Gambut Indikator Fungsi Budidaya E-G Fungsi Kawasan APL Penggunaan Lahan Pertanian Lahan Kering Pemanfaatan Hutan Rim Konseri	
		Intervensi  Sumur Bor  Rerotan Kanal  Salak Kanal  Pemampatan	
		Fungsi  1M  1H  1HK  1HT  KAWASAN	
		Kebakaran Gambut  0.1 - 1 m  1 - 2 m  2 - 3 m  3 - 5 m  5 - 7 m  > 7 m	
		Waktu Mulai 29-09-2023 08:09 Waktu Selesai 01-10-2023 17:10 Hujan Terakhir 24-08-2023 Luas Pemadaman 20.1 Ha Luas Terbakar 56 Ha Arah Angin Utara	
		Alat Pembasahan Set Mesin Pompa, Peralatan Tangan, Mobil Slip On (Pemadam) Sumber Air Kanal/Parit/Drainase, Sumur Bor	
		STATUS & TINDAK LANJUT	
		Kondisi Lahan Terbakar Tindakan Lanjut Pemadaman belum optimal Hasil Tindak Lanjut Pemadaman belum optimal Kendala Tindak Lanjut Tidak ditemukan sumber air, Musim kering air surut	

Saran/Evaluasi	penambahan unit sumur bor di lokasi rawan karhutla
-----------------------	--

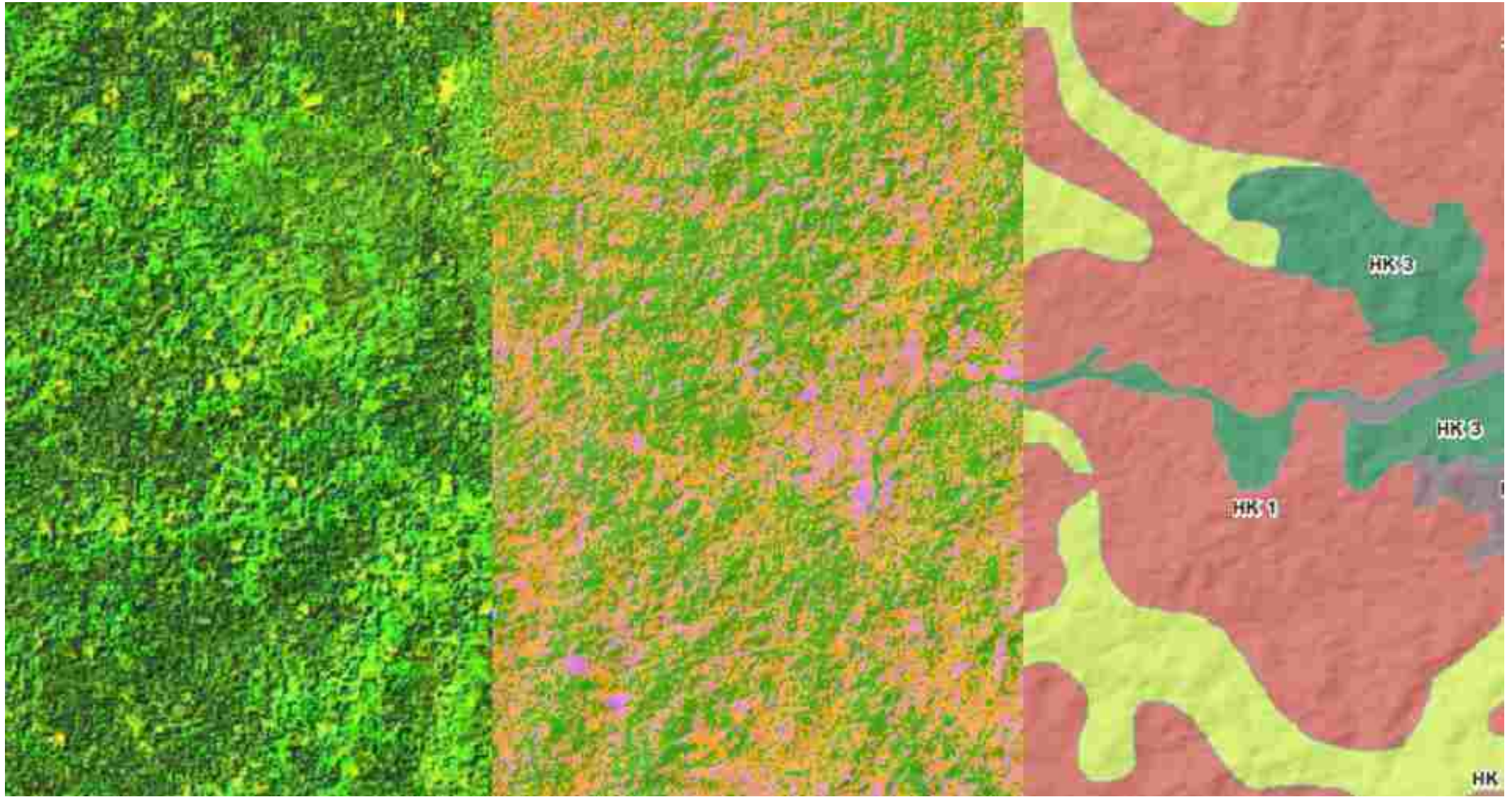
DOKUMENTASI	
 Sep, 29, 2023 2.36626069, 114.115114 Junianto Tjarko Koni Kecamatan Jabiren, Kabupaten Pulang Pisau IND	 Sep, 29, 2023 2.366267795, 114.115114 Junianto Tjarko Koni Kecamatan Jabiren, Kabupaten Pulang Pisau IND
	

Reporting

PDF generated including map, photo, signature, location-based pull data extraction. All spatially.

High Conservation Value Stratification

Map Scale 1:100,000 (2023)



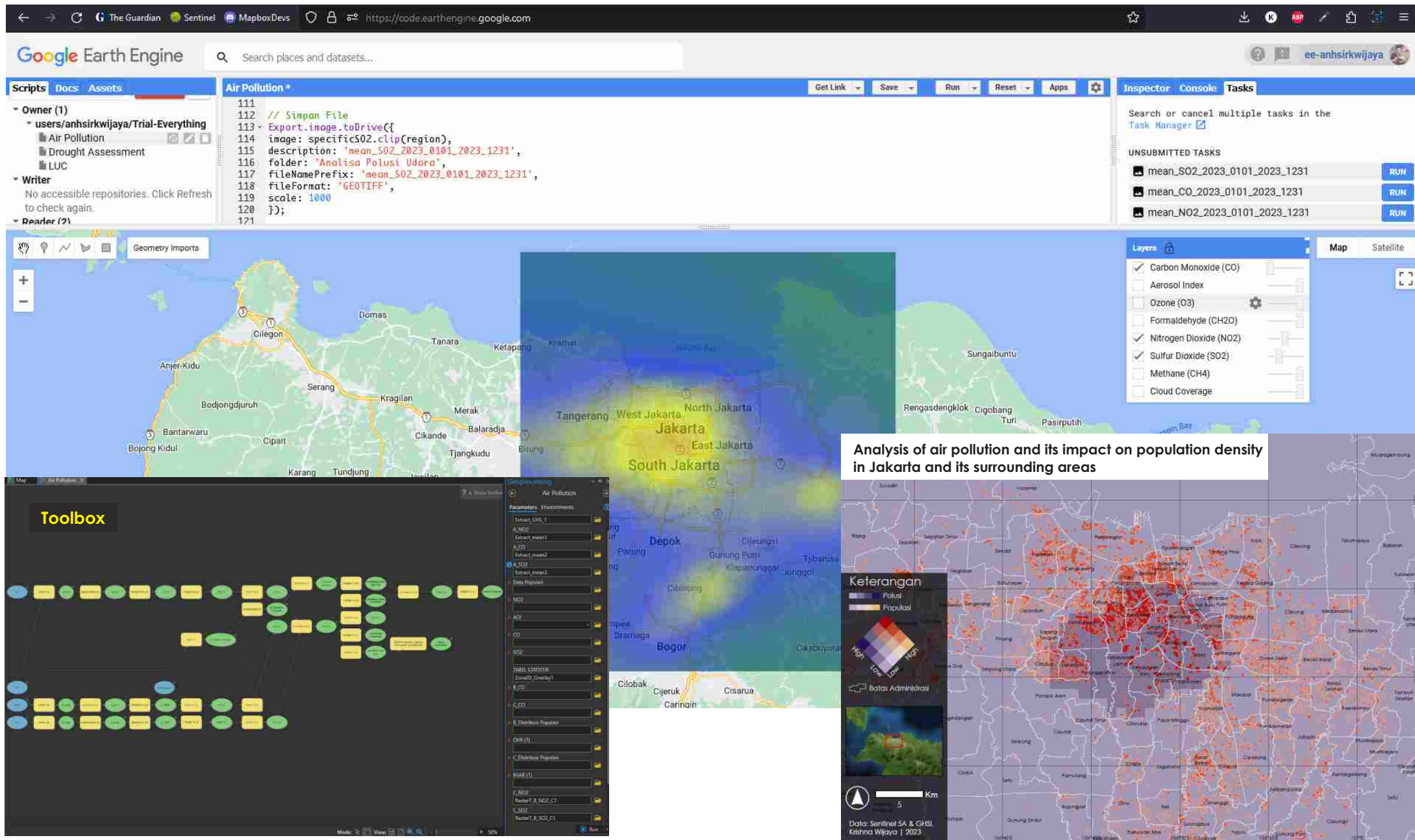
Sentinel 2
(Enhanced Vegetation Index)

Vegetation Canopy Density
(HCV-Unsupervised)

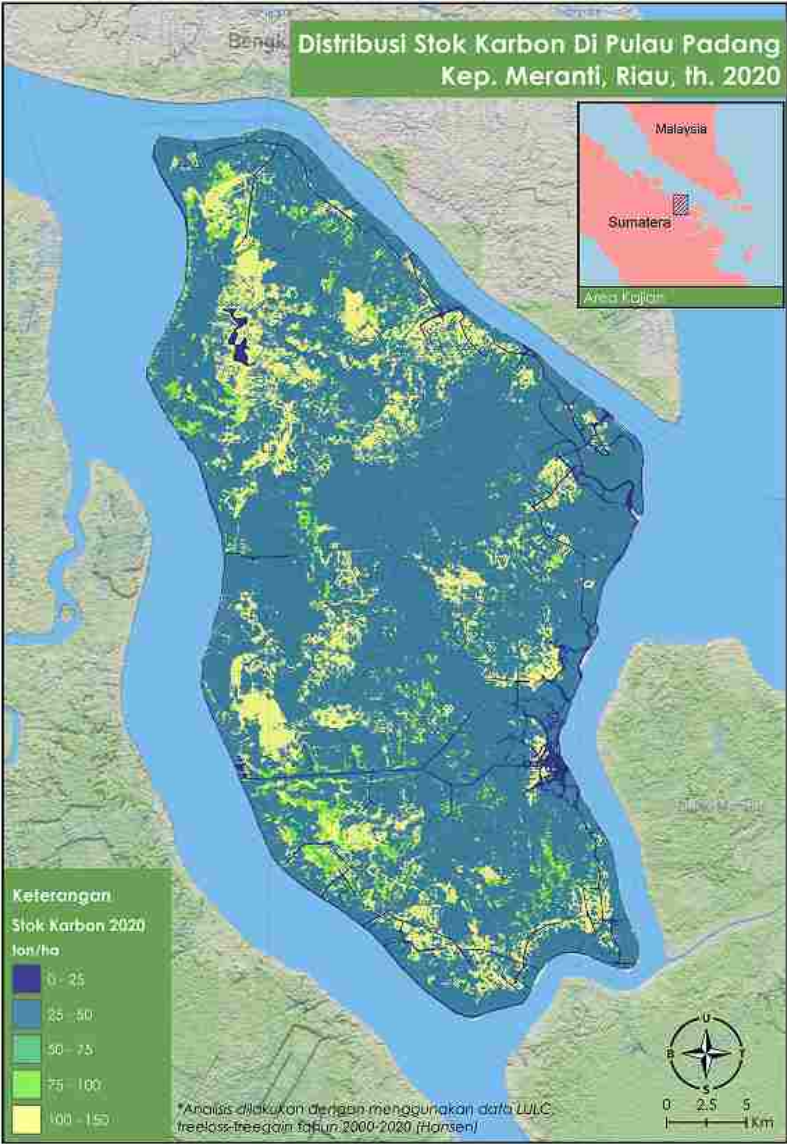
Indicative HCV
(Supervised)

Air Pollution Distribution Mapping (2023)

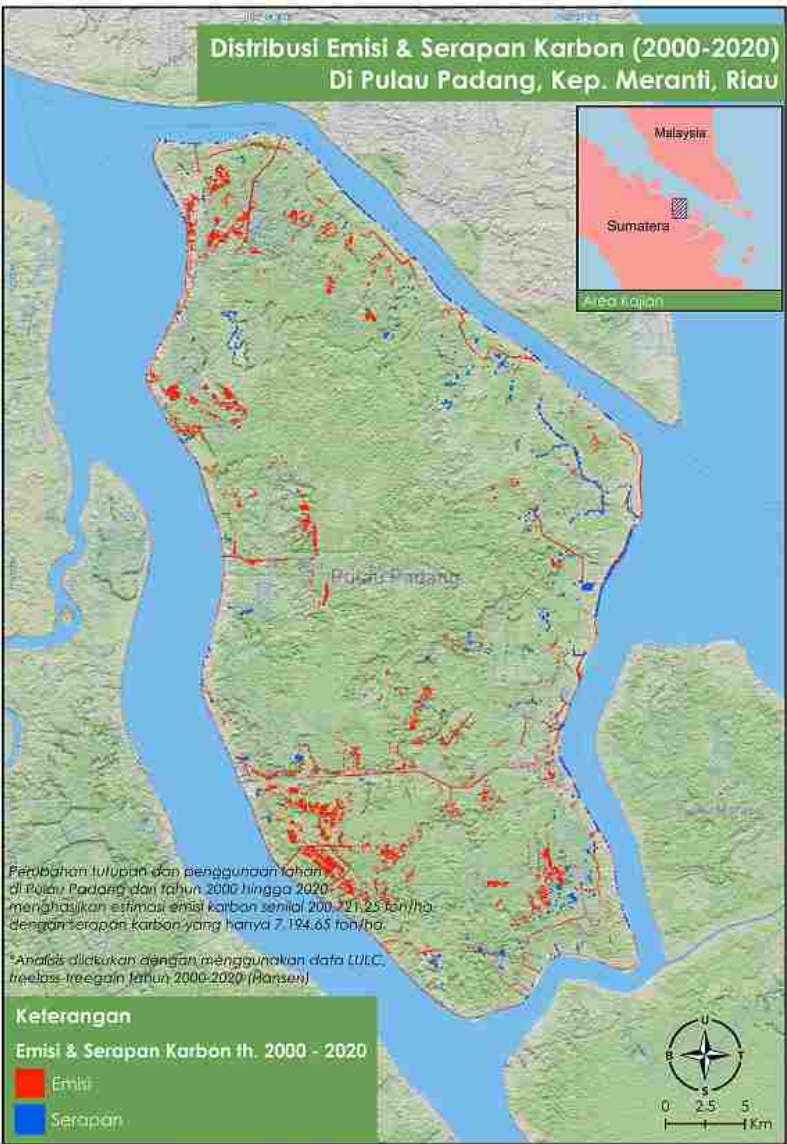
Getting insight on where air pollution impacts the population in Jakarta City



Carbon Stock Calculation (Above Ground Biomass) 2023

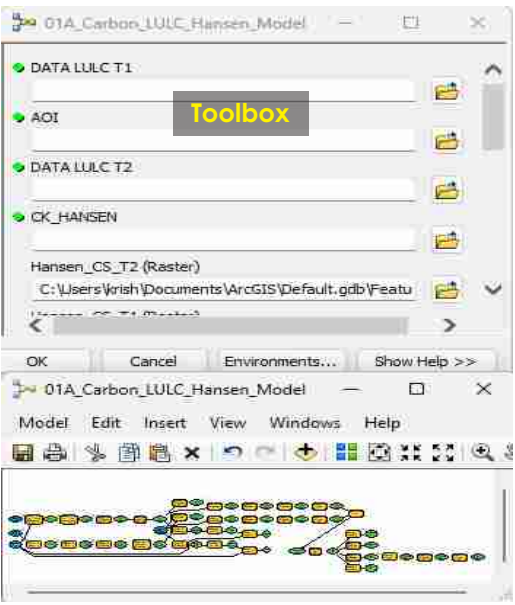


Carbon stock estimation
(baseline 2020)



Carbon stock difference
(2000-2020)

stock difference calculation based on Hansen GLULC dataset will highlight stock change between defined intervals. In this case, I used the year 2000 as the baseline with 20 years ahead interval. Since the dataset (raster) have a spatial resolution equal ~30 meters, it is best suggested for national to regional scope analysis. meanwhile KLHK developed a calibrated model that is adjusted with carbon stock measurement (constant) for each regional unit (island). Similar with the previous method, it uses annual landcover data (vector) released by KLHK at medium scale. only above ground biomass carbon measured for these 2 methods. Any specific parameter such as soil type, climate, and intensity of degradation activities are excluded.



Dipwell for Drought Monitoring (2023)

Empowering local communities to raise awareness on drought exposure in peatlands and its potential impact on crop productivity.



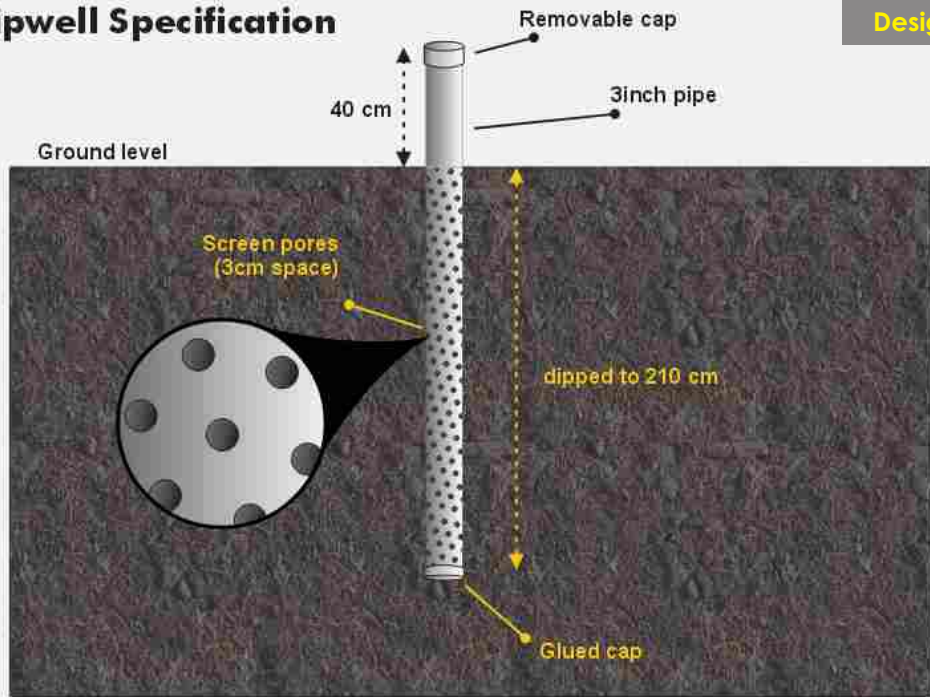
Installing



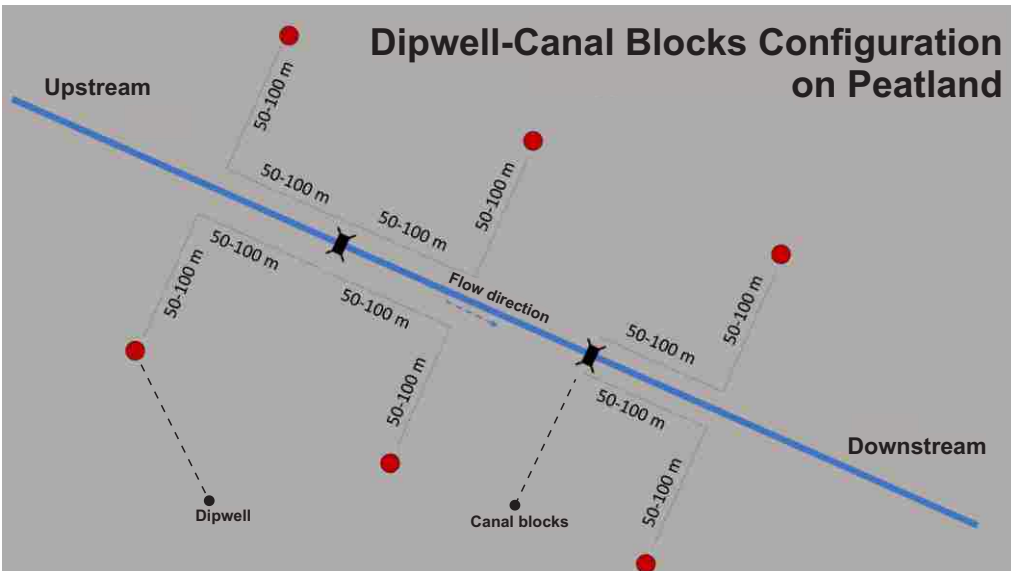
Monitoring

Dipwell for Rewetting Impact Assessment (2022)

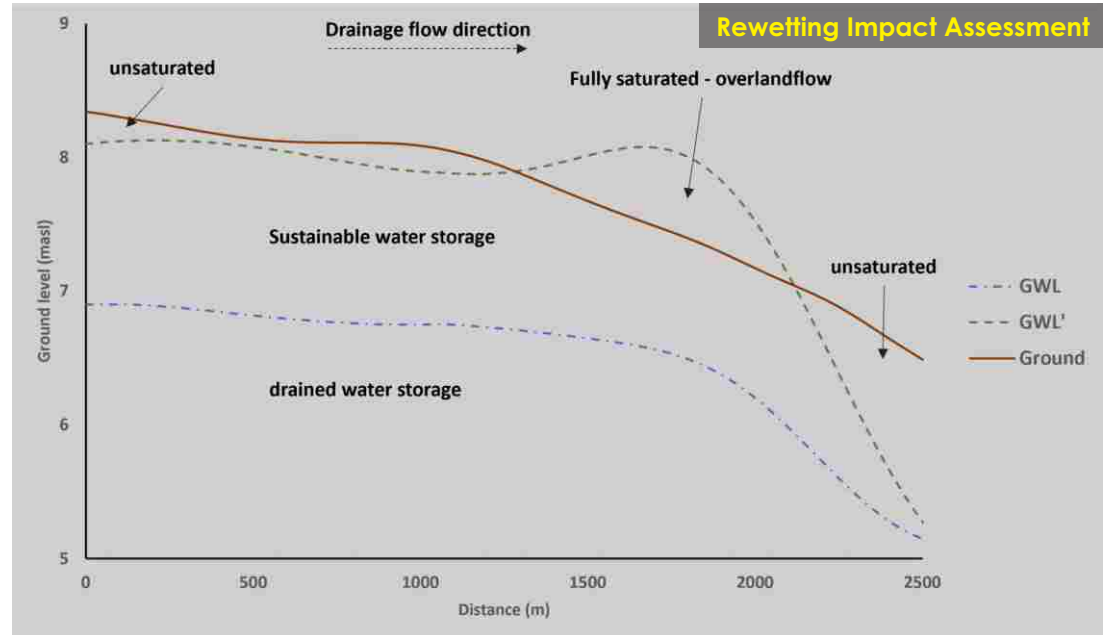
Dipwell Specification



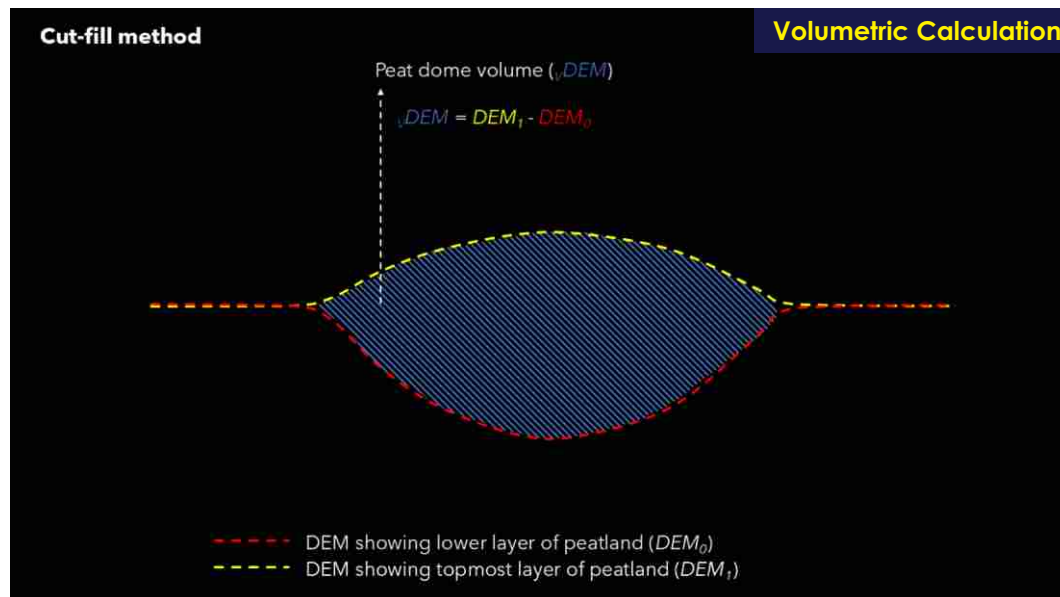
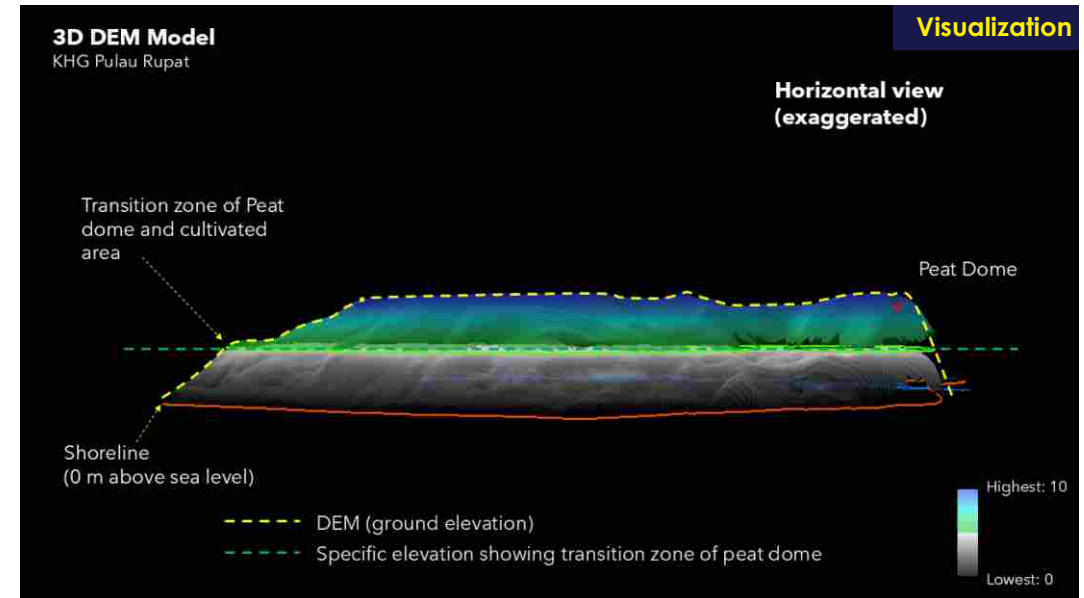
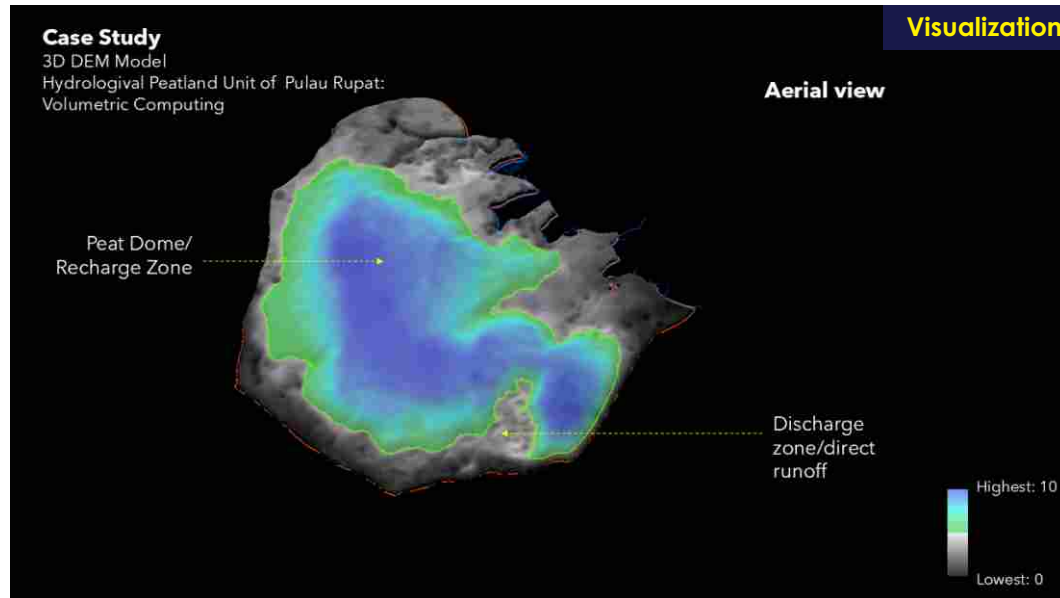
Dipwell-Canal Blocks Configuration on Peatland



Rewetting Impact Assessment



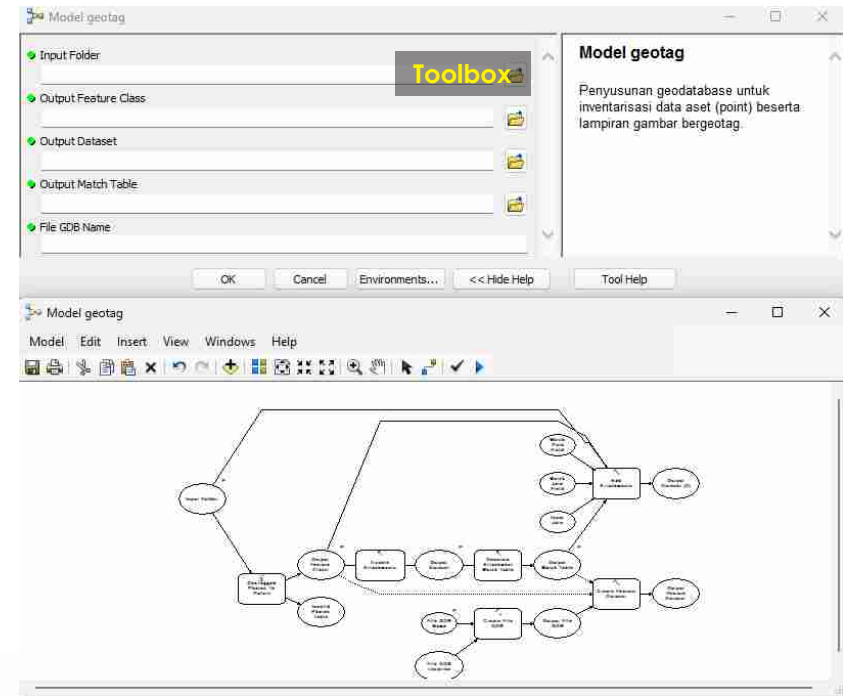
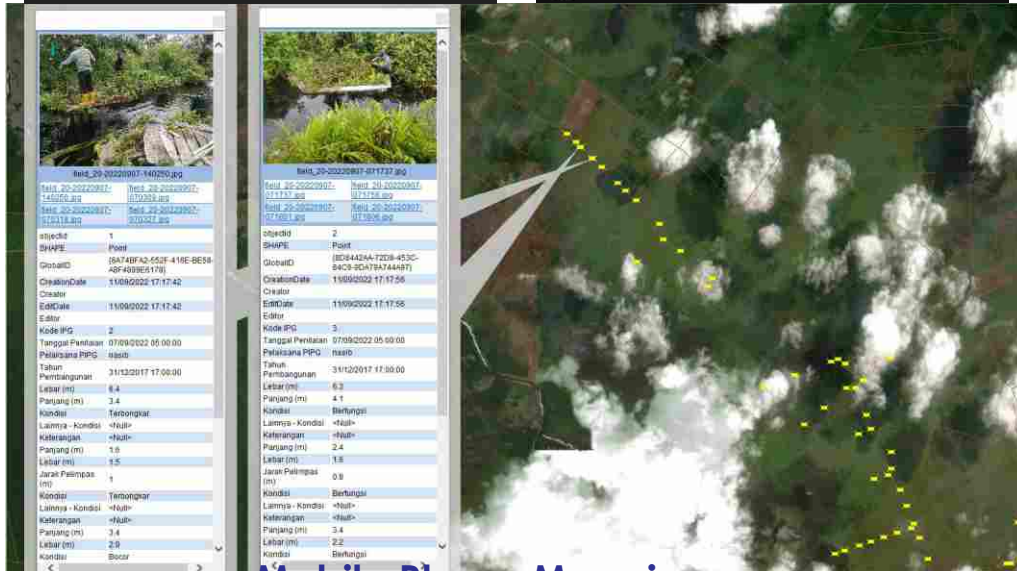
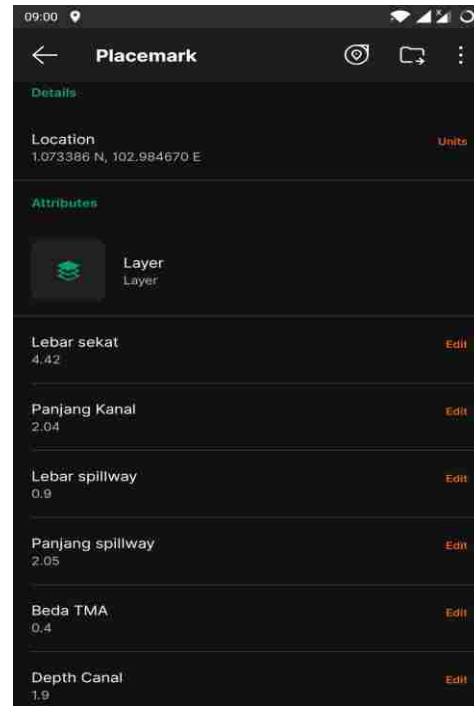
Volumetric Calculating Peat-Dome Landform (2022)



Sustainable yield (how much drained water allowed)

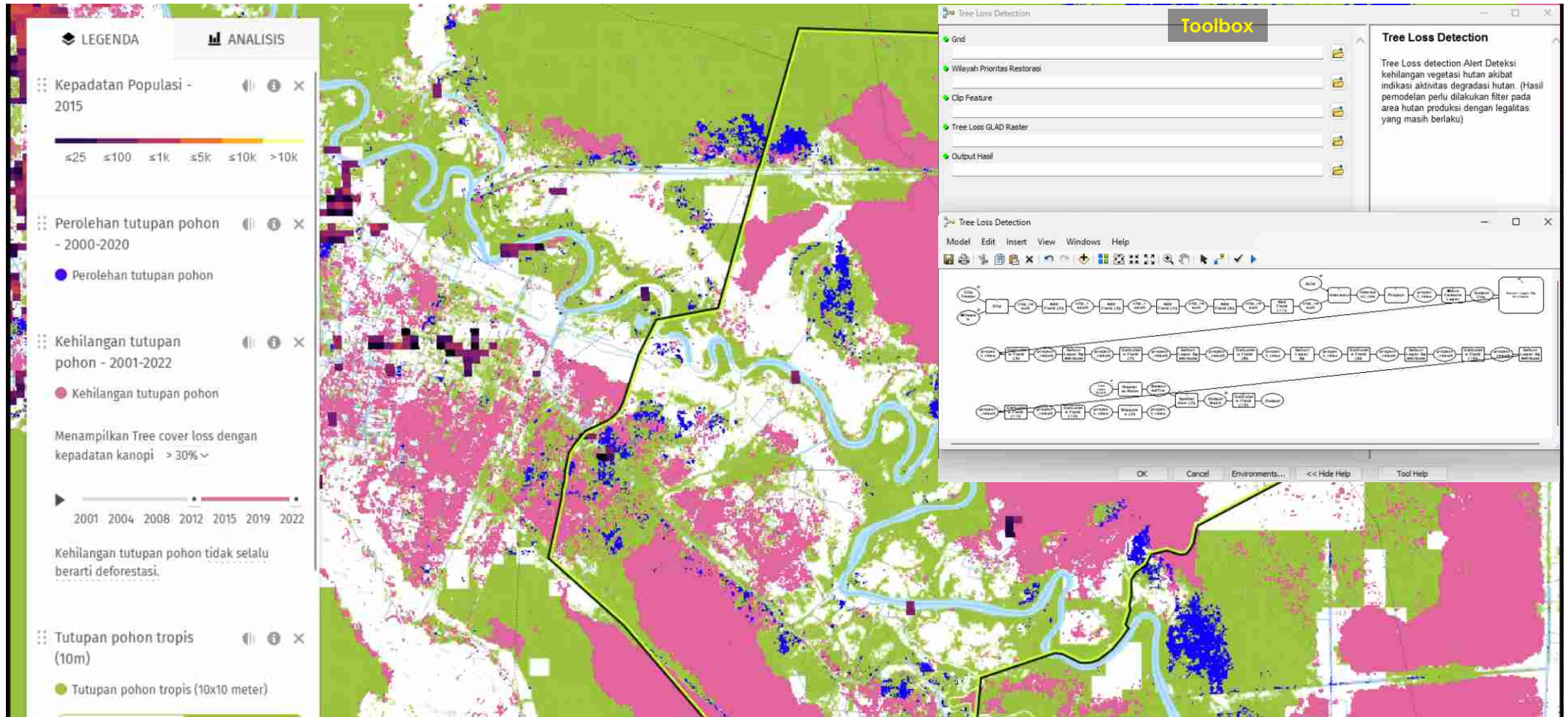
- Specific yield: volume of groundwater that can be vertically drained naturally due to gravitation.
- On peatland, water will be drained to nearby drainage or river.
- Manmade drainage aimed to accelerate the draining process due to flat slope.
- Sustainable yield can be estimated by comparing peat dome volume and annual average recharge from precipitation.

Mobile Phone Mapping, Asset Inventory, & Spatial Database Development (2021)

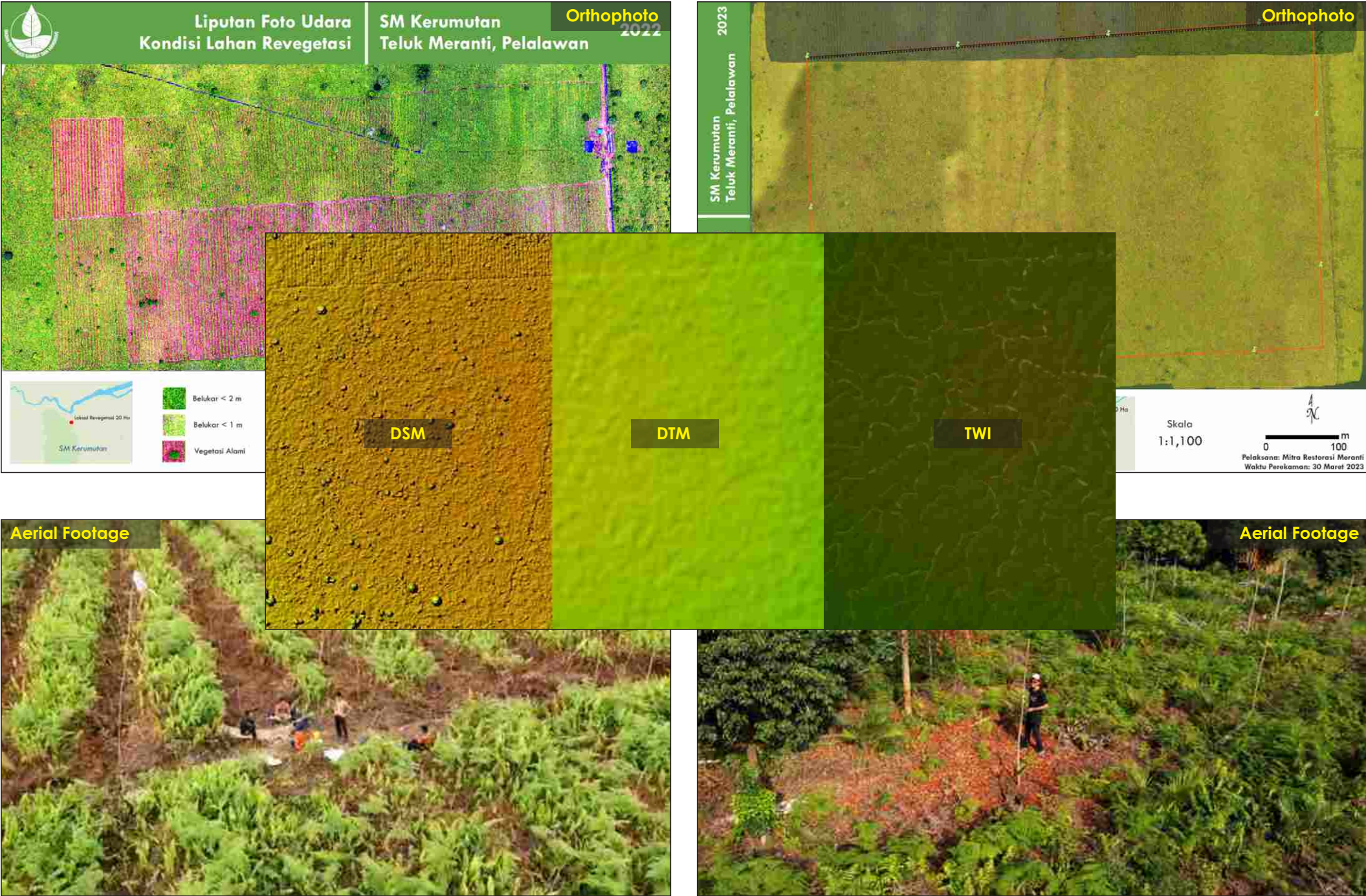


Spatial Database Development & Geotagged Picture Attachment

Forest Degradation Overview - GFW (2021-Present)

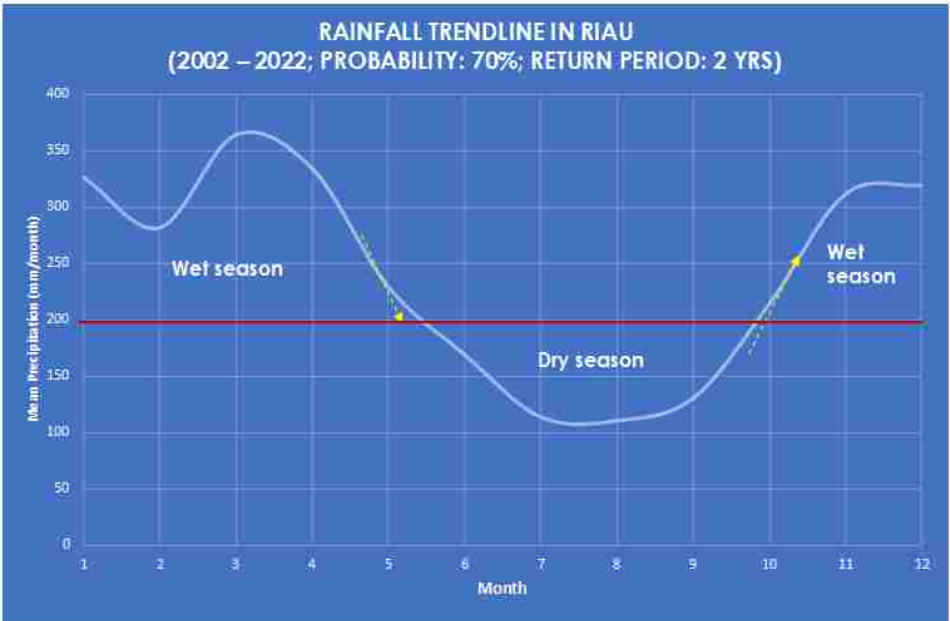


Drone Mapping for Plantation Monitoring (2020 - Present)



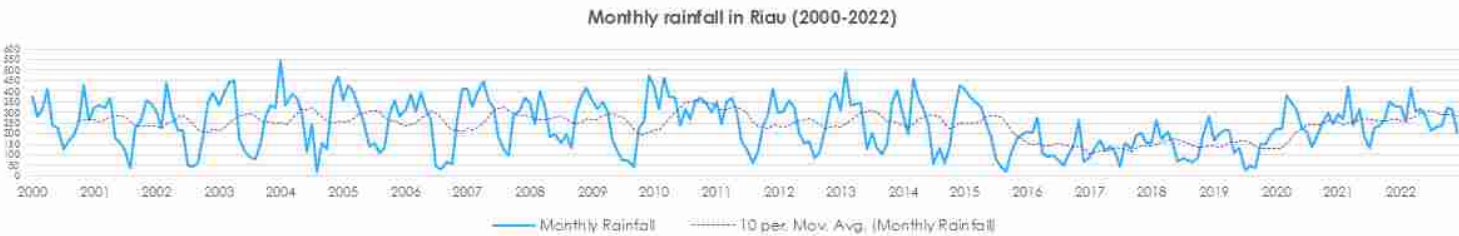
Rainfall Frequency Analysis (2001-2021)

Hydrometeorological approachment for project timeline arrangement. Ensuring successful infrastructure establishment on wetland. A better approachment to manage a suitable activities for peatland restoration, including fire risk mitigation.

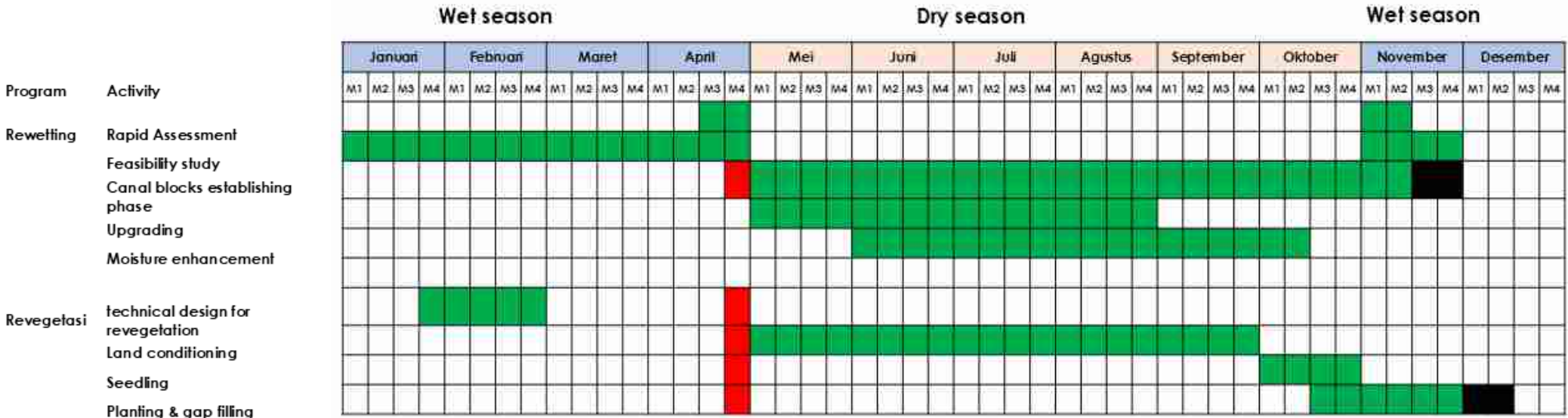


Design Rainfall Analysis for a 20-Year Period (2002–2022)
Rainfall Probability: 70% | Return Period: 2 Years
•**Dry Month Threshold:** Assumed when monthly rainfall is below 200 mm
•**Dry Season Period:** May to October
•**Wet Season Period:** November to April
•**Recommended Period for Water Construction Activities:** May to October
•**Optimal Period for Wetting Operations & Weather Modification (TMC):** June to September
•**Optimal Period for Planting & Replanting Activities:** April and October–November

*Data: GLDAS-2.1 (res. 30km)



Activity timeline for peatland restoration in Riau

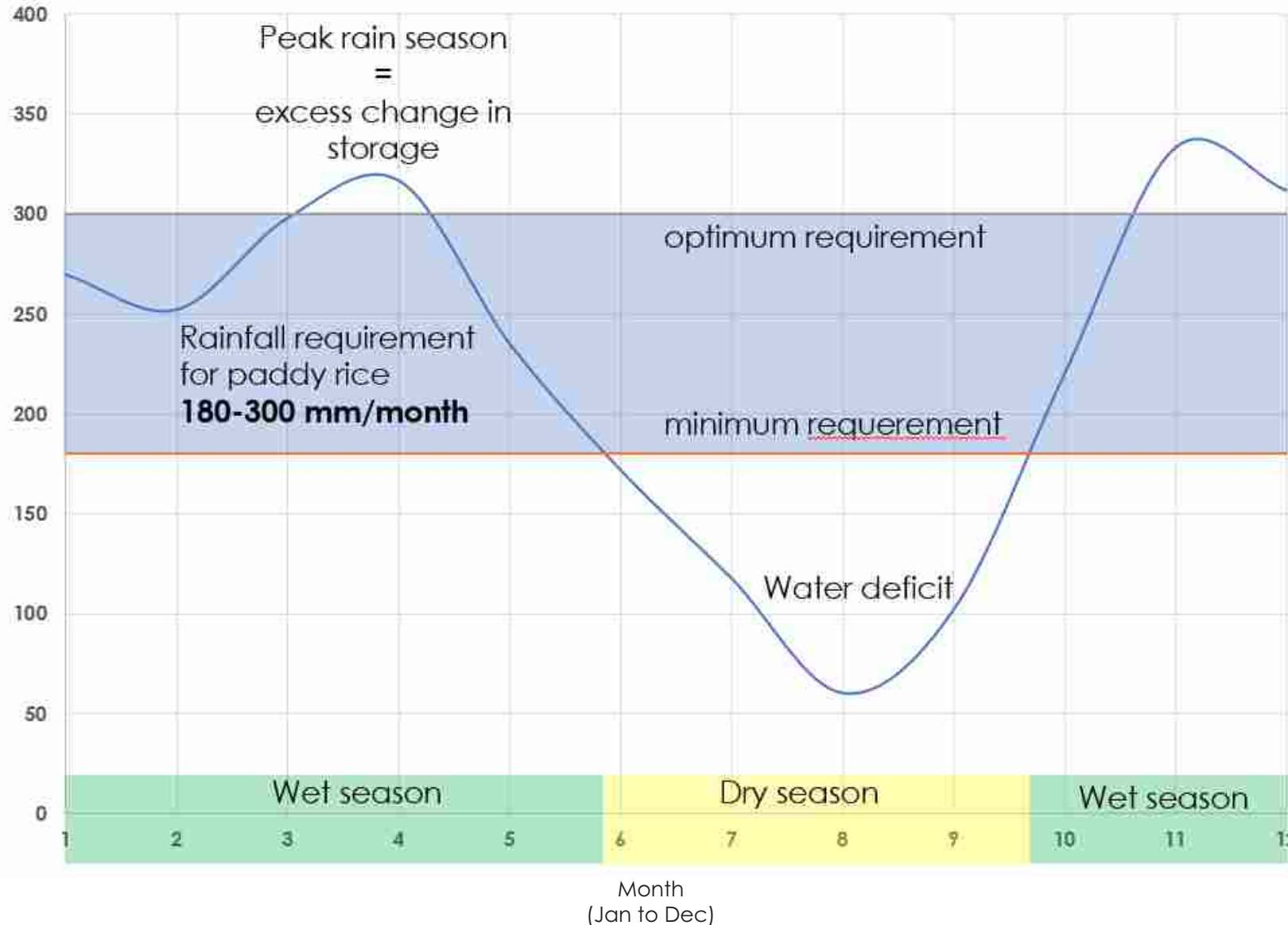


Frequency analysis of monthly rainfall (2001-2023) for paddy rice production

*Event probability: 70%

avg.
monthly
rainfall
(mm)

Monthly Rainfall in Tanjung Putri Village Kotawaringin Barat



Paddy rice typically has 3-4 months production cycle:

- Seedling stage (10-20 days)
- Vegetative stage (30-60 days)
- Reproductive stage (30-40 days)
- Maturation stage (20-30 days)

Paddy rice production cycle typically begins during wet season (October to June), which is ideal for two production cycles.

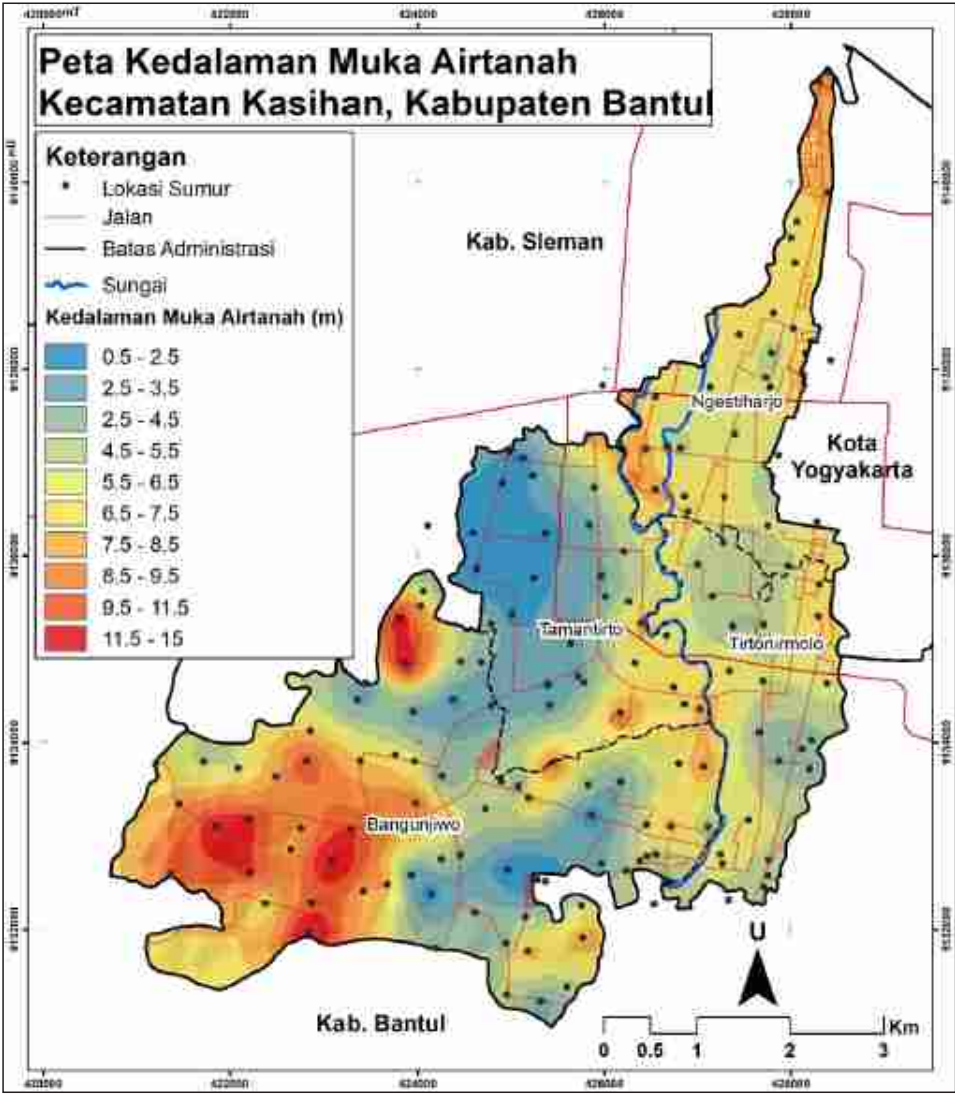
During dry season, activity should focus on optimizing rice drying, while the fields could be utilized for a short-cycle commodity crop to maximize land use.

Aerial photo captured on October 9th 2024. Showing land preparation is complete & ready for planting phase.

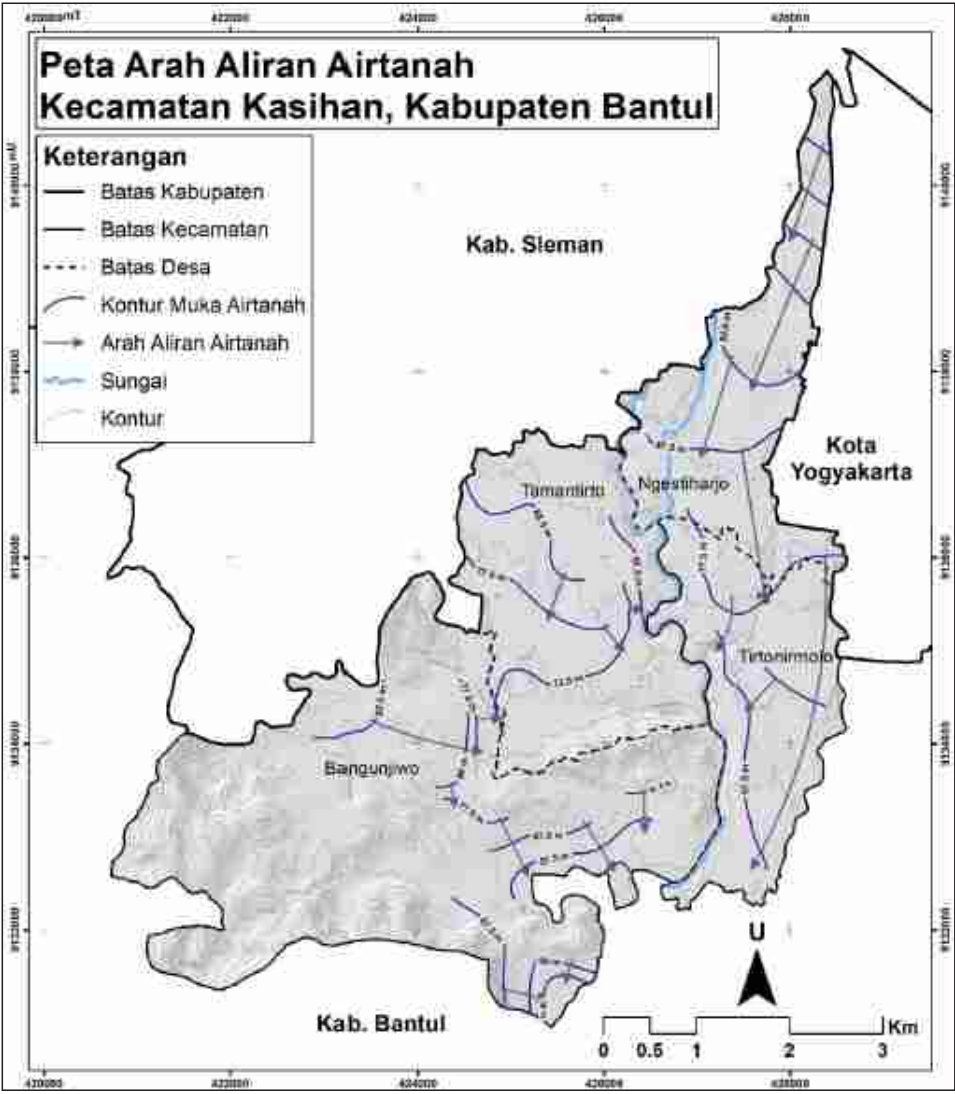


Groundwater Mapping (Thesis 2017)

Groundwater investigation for further analysis like dissolved pollutant flow, or where it prone to pollutant



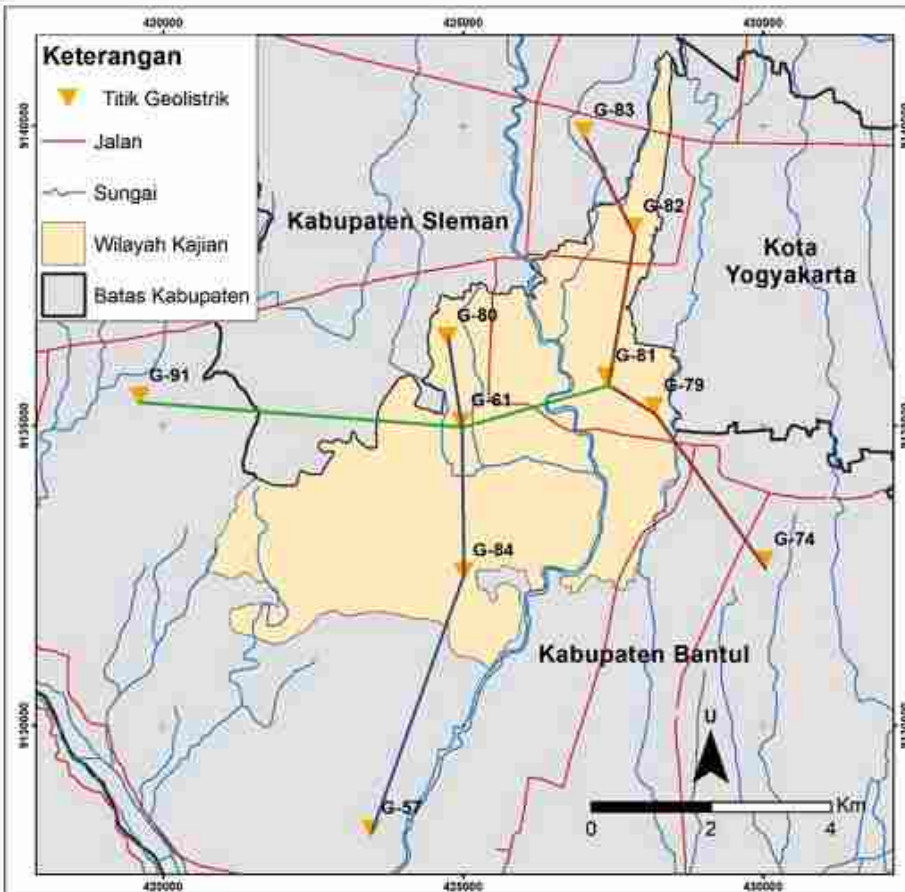
Groundwater Table Mapping



Groundwater Flownet

Potential Groundwater Investigation (Thesis 2017)

Jalur Pendugaan Geolistrik dan Model Penampang Cross Section Lapisan Bawah Tanah di Kecamatan Kasihan, Kabupaten Bantul

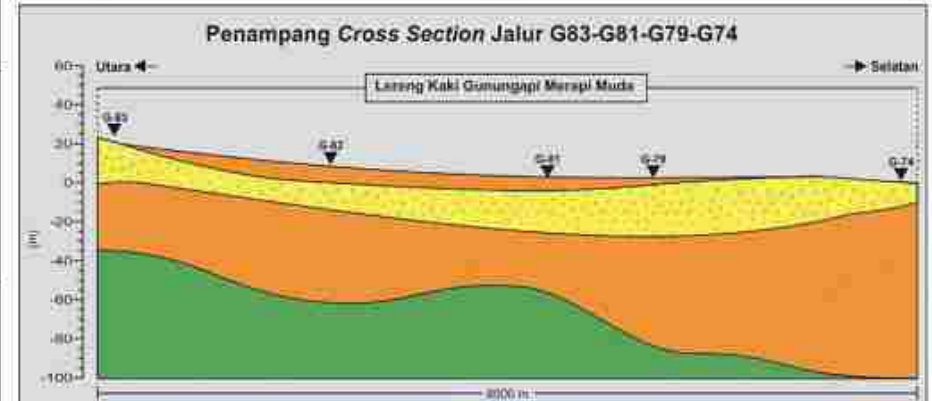
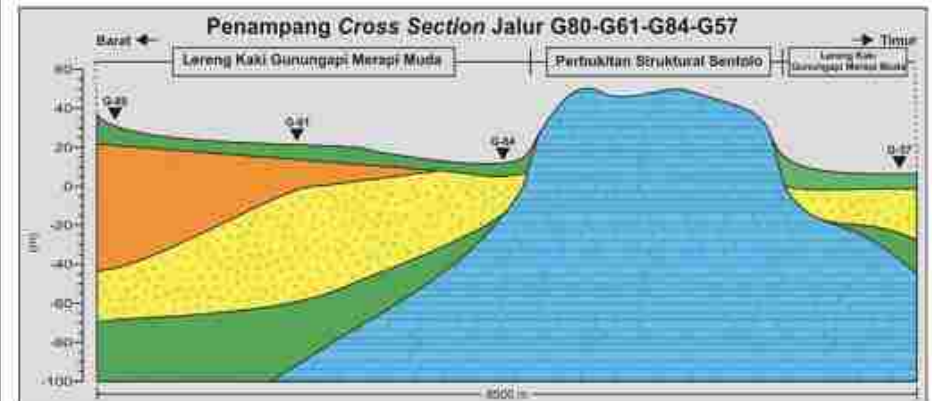
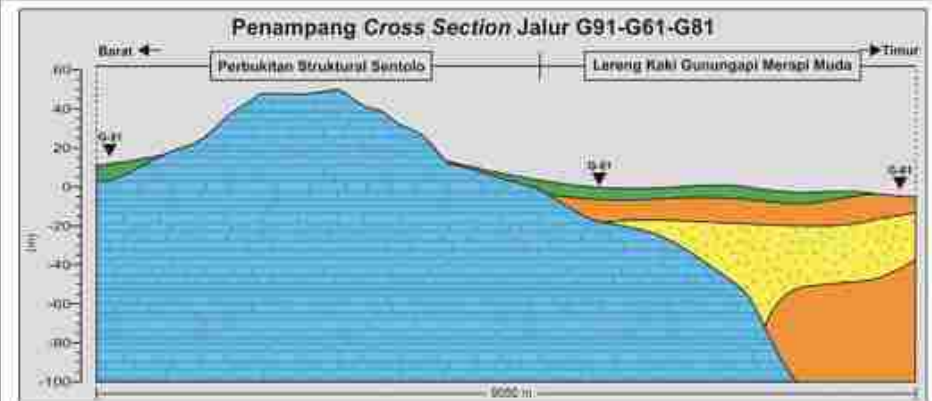


Tahanan Jenis	Simbol	Material
1-10 Ω m		Endapan lempung pada bagian atas
10-100 Ω m		Batugamping Napalan Formasi Sentolo pada bagian bawah
10-100 Ω m		Endapan pasir berlempung (pasir halus)
100-1000 Ω m		Endapan pasir dan kerikil (pasir kasar)

Sumber data: - Santosa & Adji (2014)
- Peta RBI Skala 1:25000
Lembar Yogyakarta



Dibuat oleh: Krishna Aji Wijaya
Fakultas Geografi UGM 2017

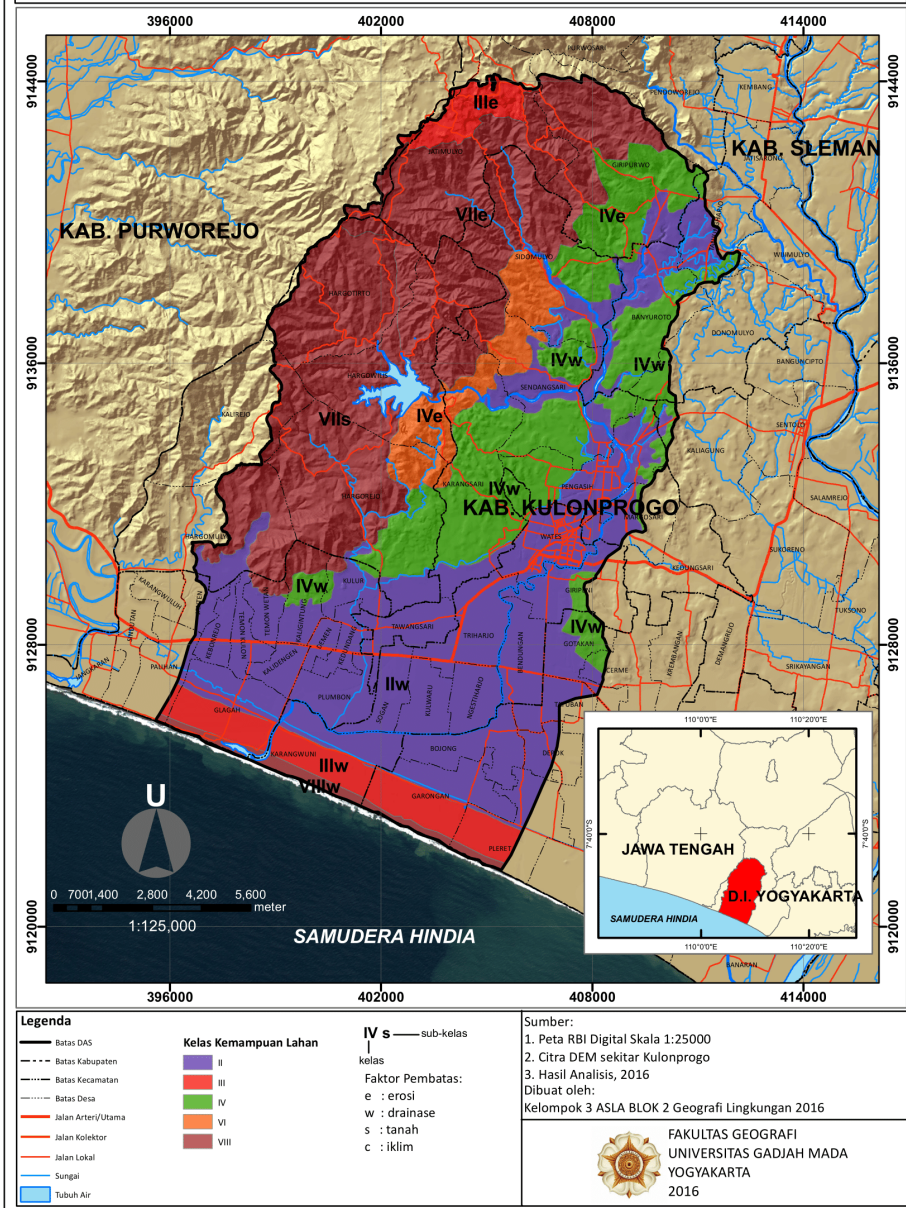


Recommended Area Development for Sustainable Crop Cultivation (2016)

Kopi robusta (*Coffea canephora*)

Persyaratan penggunaan/ karakteristik lahan	Kelas kesesuaian lahan			
	S1	S2	S3	N
Temperatur (tc) Temperatur rerata (°C)	20 - 24	24 - 28	18 - 20 28 - 32	< 18 > 32
Ketersediaan air (wa) Curah hujan (mm)	2.000 - 3.000	1.750 - 2.000 3.000 - 3.500	1.500 - 1.750 3.500 - 4.000	< 1.500 > 4.000
Lamanya masa kering (bulan)	2 - 3	3 - 5	5 - 6	> 6
Kelenturan udara (%)	45 - 80	80-90 35-45	> 90 30-35	< 30
Ketersediaan oksigen (oa) Drainase	baik	sedang	agak terhambat, agak cepat	terhambat, sangat terhambat, cepat
Media perakaran (rc) Tekstur	halus, agak halus	sedang	agak kasar	kasar, sangat halus
Bahan kasar (%)	< 15	15 - 35	35 - 60	> 60
Kedalaman tanah (cm)	> 100	75 - 100	50 - 75	< 50
Gambut: Ketebalan (cm)	< 100	100 - 200	200 - 300	> 300
Kematangan	saprik	Saprik, hemik	hemik	fabrik
Retensi hara (nr) KTK tanah (cmol)	> 16	5 - 16	< 5	
Kejenuhan basa (%)	> 20	≤ 20		
pH H ₂ O	5,3 - 6,0	6,0 - 6,5 5,0 - 5,3	> 6,5 < 5,3	
C-organik (%)	> 1,2	0,8 - 1,2	< 0,8	
Hara Tersedia (na) N total (%)	Sedang	Rendah	Sangat rendah	-
P2O5 (mg/100 g)	Tinggi	Sedang	Rendah	-
K2O (mg/100 g)	Sedang	Rendah	sangat rendah Sangat rendah	-
Toksitas (xc) Salinitas (dS/m)	< 1	-	1 - 2	> 2
Sodisitas (xn) Alkalinitas/ESP (%)	-	-	-	-
Bahaya sulfidik (xs) Kedalaman sulfidik (cm)	> 175	125 - 175	75 - 125	< 75
Bahaya erosi (eh) Lereng (%)	< 8	8 - 15	15 - 30	> 30
Bahaya erosi	sangat ringan	Ringan - sedang	berat	sangat berat
Bahaya banjir/genangan pada masa tanam (fh) - Tinggi (cm)	-	-	25	> 25
- Lama (hari)	-	-	< 7	≥ 7
Penyiapan lahan (lp) Batuan di permukaan (%)	< 5	5 - 15	15 - 40	> 40
Sinkapan batuan (%)	< 5	5 - 15	15 - 25	> 25

PETA KEMAMPUAN LAHAN METODE SUBJECTIVE MATCHING DAS SERANG, KULONPROGO





ArcGIS



ArcGIS Pro



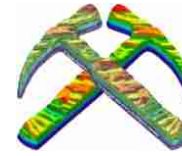
Agisoft
Metashape



Pix4Dcapture



ArcGIS Survey123



Surfer

GlobalMapper

QGIS



ENVI



ArcGIS
Online



AUTOCAD
MAP 3D



Epicollect5



CapCut



Corel DRAW
Graphics Suite



DroneDeploy



AVENZA MAPS



mapbox